DRUGS · BLEEDING AND HAEMOSTASIS

Topical haemostatic agents

Topical haemostatic agents stop bleeding where pressure, diathermy and ties are impractical: diffuse oozing in the nose and sinuses, bone surfaces, the skull base, tonsillar beds and the sigmoid sinus in ear surgery. They work in different ways, and choosing the right one depends on whether you need a passive scaffold, an active clotting agent, or a sealant.

1 CHAPTER ONE At a glance

The agents on one page.

1.1 Summary

TOPICAL HAEMOSTATIC AGENTS

	SUMMARY
Passive scaffolds	Oxidised regenerated cellulose (Surgicel); gelatin sponge (Gelfoam, Spongostan); microfibrillar collagen; bone wax. They give a framework for platelets and clot; oxidised cellulose is also acidic and mildly antibacterial
Active agents	Thrombin preparations; gelatin-thrombin matrices (Floseal, Surgiflo with thrombin): flowable gels that fill irregular bleeding surfaces
Sealants	Fibrin sealants (fibrinogen and thrombin, such as Tisseel); synthetic polyethylene glycol sealants. Used for haemostasis and to seal CSF leaks and grafts
Absorbable packs	Hyaluronic acid and carboxymethylcellulose packs (nasal), which hold the mucosa and dissolve
ENT uses	Epistaxis in patients on anticoagulants or with bleeding disorders (absorbable packs or matrices, avoiding packs that need removal); endoscopic sinus and skull base surgery; tonsil bed oozing; sigmoid sinus and emissary vein bleeding; bone bleeding (wax)

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Choosing

- **Oozing from a sinus cavity**: a gelatin-thrombin matrix applied directly and left in place, or an absorbable pack.
- **Sigmoid sinus tear in mastoid surgery**: oxidised cellulose or gelatin sponge held with a pledget and gentle pressure, then left; do not pack into the lumen.
- **Bone bleeding** (emissary veins, cut marrow): bone wax, used sparingly because it inhibits bone healing and is a foreign body.
- **Skull base repair**: fibrin or synthetic sealant over the graft layers, not between them.

2.2 In epistaxis

Absorbable products avoid the trauma and rebleeding of pack removal, especially in patients with hereditary haemorrhagic telangiectasia and coagulopathy.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- Swelling of oxidised cellulose and gelatin, which can compress nerves in confined spaces (optic canal, foramina, spinal canal): remove excess in such sites.
- Foreign body reaction and granuloma; infection; adhesions.
- Bovine thrombin can induce antibodies against factor V; human or recombinant thrombin avoids this.
- Thrombin products must **never enter the circulation** (fatal thrombosis): do not inject into vessels.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The difference between **passive, active and sealant** agents.
- Managing a sigmoid sinus bleed.
- Risks of swelling and intravascular thrombin.

4.2 Common pitfalls

COMMON PITFALLS

- Leaving swelling haemostats around the optic nerve.
- Overusing bone wax.

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